

Introduction

The light emitted by a gas lamp corresponds to the energy differences of different excitation states of electron in the atoms. The different wavelengths of light emitted by an element form the spectrum for that atom.

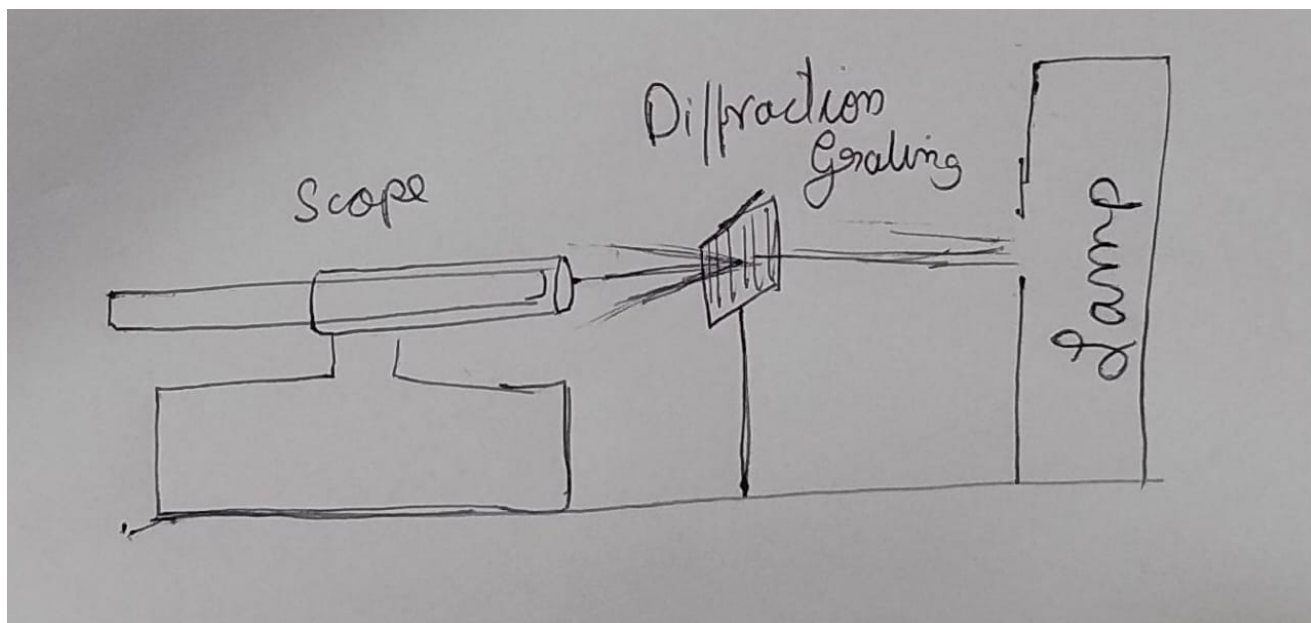
The goal of this experiment is to determine the spectrum of light emitted by the Mercury Vapor Lamp.

Experiment Setup and Procedure

Apparatus

Sodium Vapor Lamp, Mercury Vapor lamp, Diffraction grating, *scope with provision for measuring angles*.

Experiment Setup



Procedure

First, the sodium lamp is used. Through the telescope, the angle at which light is diffracted by the grating is measured. The wavelength of light emitted by sodium lamp is known. Using this information, the spacing in the lines of the grating is calculated.

Now, the Mercury lamp is put in its place. The grating diffracts different wavelengths at different angles. Through the telescope, the diffraction angles for different colors of light emitted by the lamp are measured. As the spacing of grating was calculated in the first part of the experiment, using the data of diffraction angles, the different wavelengths can be calculated and so the emission spectrum of mercury lamp can be determined.

Formulae and Relations

Maxima's by diffraction grating:

$$n \lambda = d \sin(\theta)$$

n = order of maxima

λ = wavelength of light

d = spacing of the diffraction grating

θ = angle of maxima

We only observe the first order maxima in this experiment. So, $n = 1$

In the first part of the experiment, we get

$$d = \frac{\lambda}{\sin(\theta)}$$

In the second part, using d , we determine

$$\lambda = d \sin(\theta)$$

Data and Analysis

The sodium vapor lamp emits two close wavelengths. We consider there average in the is experiment.

$$\lambda_0 = 589.3 \text{ nm}$$

The following readings were taken for maxima's of sodium lamp

1st Order Maxima	Central Maxima	1st Order Maxima
218°	239.5°	260.5°

$$\therefore \theta_0 = \frac{260-218}{2} = 21.25^\circ$$

So, we get

$$d = \frac{\lambda_0}{\sin(\theta_0)} = 1625.93 \text{ nm}$$

Now, replacing the Sodium vapor lamp with the Mercury vapor lamp, the following readings for angles were taken.

	1st Order Maxima(θ_1)	1st Order Maxima(θ_2)	$\theta = \frac{\theta_2 - \theta_1}{2}$	$\lambda = d \sin(\theta)$
Violet	225.5°	254°	14.25°	400.22 nm
Blue	224.5°	255°	15.25°	427.67 nm
Cyan	222°	257°	17.5°	482.16 nm
Green	220.5°	259°	19.25°	536.05 nm
Yellow	219°	260°	20.5°	569.41 nm

Thus the spectrum of the mercury vapor lamp has been determined.